P1.8 Electrical appliances

Learning objectives

After this topic, you should know:

- how energy is supplied to your home
- why electrical appliances are so useful
- what most everyday electrical appliances are used for
- how to choose an electrical appliance for a particular job.

Synoptic links

You should know that electricity is not an energy store, but rather it is a flow of charge that transfers energy from one energy store to another. You will learn more about electricty in Chapter P4.

Everyday electrical appliances

The energy you use in your home is mostly supplied by electricity, gas, and oil. Although all three of these energy supplies can be used for cooking and heating, your electricity supply is vital because you use electrical appliances for so many purposes every day. The charge that flows through these appliances transfers energy to them, which they then transfer usefully. But some of the energy you supply to them is wasted.

Figure 1 *Electrical appliances – how many can you see in this photo?*

Table 1 *Comparing energy use in electrical appliances*

Appliance	Useful energy	Energy wasted
Light bulb	Light emitted from the glowing filament.	Energy transfer from the filament heating the surroundings.
Electric heater	Energy heating the surroundings.	Light emitted from the glowing element.
Electric toaster	Energy heating bread.	Energy heating the toaster case and the air around it.
Electric kettle	Energy heating water.	Energy heating the kettle itself.
Hairdryer	Kinetic energy of the air driven by the fan. Energy heating air flowing past the heater filament.	Sound of fan motor (energy heating the motor heats the air going past it, so is not wasted). Energy heating the hairdryer itself.
Electric motor	Kinetic energy of objects driven by the motor. Gravitational potential energy of objects lifted by the motor.	Energy heating the motor and energy transferred by the sound waves generated by the motor.

Clockwork radio

People without electricity supplies can now listen to radio programmes – thanks to the British inventor Trevor Baylis. In the early 1990s, he invented the clockwork radio. When you turn a handle on a clockwork radio, you wind up a clockwork spring in it, and increase the elastic potential energy of the spring. When the spring unwinds, energy from its elastic potential energy store is transferred to its kinetic energy store and it turns a small electric generator in the radio. It doesn't need batteries or mains electricity. So people in remote areas where there is no mains electricity can listen to their radios without having to walk miles for a replacement battery. But they do have to wind up the spring every time its store of energy has been used.

Figure 2 *Clockwork radios are now mass produced and sold all over the world*

Choosing an electrical appliance

You use electrical appliances for many purposes. Each appliance is designed for a specific purpose, and it should waste as little energy as possible. Suppose you were a rock musician at a concert. You would need appliances that transfer the variations in sound waves to electricity and then back to sound waves. But you wouldn't want the appliances to transfer lots of energy to the thermal energy store of the surroundings or themselves. See if you can spot some of these appliances in Figure 3.

Figure 3 *On stage*

1 Match each electrical appliance in the list below with the energy transfer A or B it is designed to bring about.

Energy transfer **A** Energy transferred by an electric current → energy transferred by sound waves

B Energy transferred by an electric current → kinetic energy store

a Electric drill [1 mark]
b Food mixer [1 mark]
c Electric bell [1 mark]

2 **a** Explain why a clockwork radio needs to be wound up before it can be used. [2 marks]
b Describe the changes in energy stores that take place in a clockwork radio when it is wound up and then switched on. [2 marks]
c Give one advantage and one disadvantage of a clockwork radio compared with a battery-operated radio. [2 marks]

3 An electric dishwasher heats water and sprays the hot water at the dishes. It then pumps the water out.
a Describe how energy is usefully used in the machine. [1 mark]
b Describe how energy is wasted by the machine. [2 marks]

4 A laptop battery stores energy with an efficiency of 80% when it is recharged from a low-voltage power supply. When it is connected to the laptop, it transfers energy to the laptop with an efficiency of 60%.
a Calculate the overall percentage efficiency of the laptop battery. [2 marks]
b Calculate the overall percentage of energy wasted. [1 mark]

Key points

- Electricity and gas and/or oil supply most of the energy you use in your home.
- Electrical appliances can transfer energy in the form of useful energy at the flick of a switch.
- Uses of everyday electrical appliances include heating, lighting, making objects move (using an electric motor), and producing sound and visual images.
- An electrical appliance is designed for a particular purpose and should waste as little energy as possible.